

ADITYA KARTHIK SARAVANAKUMAR

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RESEARCH INTERESTS

Computational Fluid Dynamics Finite Element Modeling High-Performance Computing Machine Learning

EDUCATION

Doctor of Philosophy, Massachusetts Institute of Technology

Major: Scientific Computing and Mechanical Engineering

Minor: Machine Learning

May 2023 - Present

GPA: 5.00/5.00

AWARDS

MIT MathWorks 2025 Fellowship

Graduate fellowship supporting computational science research

Master of Science, Massachusetts Institute of Technology

Major: Computational Science and Engineering

Sep 2021 - May 2023

GPA: 5.00/5.00

DAAD WISE 2020 Fellowship

Fully-funded 12-week internship at RWTH Aachen University

Bachelor of Engineering, Birla Institute of Technology & Science

Major: Mechanical Engineering

Sep 2017 - May 2021

GPA: 9.78/10.00

PROGRAMMING LANGUAGES AND SIMULATION FRAMEWORKS

C++ Python PyTorch git OpenMP MPI deal.II ANSYS Fluent COMSOL FEniCS OpenFOAM

RESEARCH PROJECTS

Development of an Adaptive High-Order Locally Non-Hydrostatic Ocean Model

Graduate Research Project | Guide: Prof. Pierre Lermusiaux | MSEAS Lab

Massachusetts Institute of Technology

Sep 2021 - Ongoing

Designed a novel ocean model that locally resolves complex dynamics and achieves computational savings compared to traditional globally-non-hydrostatic models. Currently incorporating this model into a C++ hybridizable Discontinuous Galerkin solver and devising numerical techniques to overcome computational limitations and conduct large-scale regional ocean simulations.

Machine Learning and Stochastic Modeling Projects

Massachusetts Institute of Technology

Deep Learning: Vision transformer models, optimized for GPUs, to predict large multi-scale time-series fluid dynamics data.

Numerical Methods for Stochastic PDE: Multilevel Monte Carlo methods for variance reduction in stochastic elliptic PDE solutions.

Reinforcement Learning: Deep reinforcement learning models to precondition linear systems arising from PDE discretization.

Analysis and Extension of a Rarefied Gas Dynamics Solver

RWTH Aachen University

Bachelor Thesis | Guide: Prof. Manuel Torrilhon

May 2020 - Jan 2021

Studied the stability of a FEniCS-based Discontinuous Galerkin FEM solver that solves the linearized regularized 13 (R13) moment equations to study non-equilibrium gas dynamics problems. Extended solver to accommodate non-linear conservation equations.

Numerical Investigation of Flapping Dynamics of Multi-Layered Plates

Birla Institute of Technology and Science

Undergraduate Research Project | Guide: Prof. Pardha S. Gurugubelli

Feb 2020 - May 2021

Employed a MATLAB-based finite element method code to study fluid-structure interaction dynamics of multi-layered plates flapping in a uniform flow. Enhanced solver to allow coupling between high-order fluid elements and low-order structural elements.

Detached Eddy Simulations of Flow Past Cylinder at High Reynolds Numbers

Birla Institute of Technology and Science

Undergraduate Research Project | Guide: Prof. Sabareesh G. Rajasekharan

Aug 2019 - Jan 2021

Optimized building design for the LIGO India observatory to minimize wind load and vibrations through corner modification and windbreaks. Executed 3D Detached Eddy Simulations on ANSYS FLUENT to capture turbulence at high Reynolds number regimes.

INDUSTRY EXPERIENCE

Atomistic Modeling of Deposition Mechanisms in Semiconductor Manufacturing Simulations

Lam Research

Computational Science Software Engineering Intern | Guide: Dr. Raghuraj Hathwar

Jun 2025 - Aug 2025

Developed theoretical models for deposition mechanisms in semiconductor manufacturing, including nucleation delay, ion-assisted deposition, and ion densification. Implemented and validated these features within the C++ SEMulator3D codebase, allowing for efficient higher-fidelity process simulations.

Magnetohydrodynamic Simulations of Flow in Induction Pumps

Indira Gandhi Center for Atomic Research

Research Intern (Thermal Hydraulics Division) | Guide: Dr. Kumaresan Natesan

May 2019 - Jul 2019

Simulated MHD flows by coupling Navier-Stokes and Maxwell's equations in COMSOL. Validated the method with Hunt's benchmark laminar solution and conducted preliminary simulations for Annular Linear Induction Pumps used in fast breeder reactors.

PUBLICATIONS

Saravanakumar, A. K., C. Foucart, C. Mirabito, P. J. Haley Jr., and P. F. J. Lermusiaux (2025). "An Adaptive High-Order Locally-Non-Hydrostatic Ocean Model". In: To be submitted to Ocean Modeling.

Saravanakumar, A. K. (2023). "Towards Coupled Nonhydrostatic-Hydrostatic Hybridizable Discontinuous Galerkin Method". SM thesis. Massachusetts Institute of Technology, Computational Science and Engineering.

Saravanakumar, A. K., K. Supradeepan, et al. (2021). "A numerical study on flapping dynamics of a flexible two-layered plate in a uniform flow". In: *Physics of Fluids* 33.1, p. 017108. ISSN: 1070-6631. DOI: 10.1063/5.0033049.

Vishwanath, N., A. K. **Saravanakumar**, et al. (2022). "3D Numerical investigation of a rounded corner square cylinder for supercritical flows". In: *Wind and Structures* 35.1, pp. 55–66. ISSN: 1226-6116. DOI: 10.12989/was.2022.35.1.055.

Brahmini Priya, P. V., A. K. **Saravanakumar**, et al. (2023). "Numerical Analysis of Two-Degrees of Freedom Vortex-Induced Vibrations of an Elastically Mounted Circular Cylinder Placed Very Close to a Wall". In: *Sub-Judice*. DOI: 10.2139/ssrn.4470199.

PRESENTATIONS

Saravanakumar, A. K., C. Foucart, C. Mirabito, P. J. Haley, and P. F. Lermusiaux (2024a). "Adaptive Nonhydrostatic-Hydrostatic Hybridizable Discontinuous Galerkin Ocean Solver". In: *Ocean Sciences Meeting, New Orleans, Louisiana, 2024*. AGU.

Saravanakumar, A. K., C. Foucart, C. Mirabito, P. J. Haley, and P. F. Lermusiaux (2024b). "An Adaptive High Order Locally-Nonhydrostatic Ocean Solver". In: *Mathematics of Planet Earth, Portland, Oregon, 2024*. SIAM.